

OldPhoneDeals eCommerce Web Application

Presented by Tutorial 8 - Group 5
Faculty of Engineering,
COMP5347 Group Based Assignment



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Project Overview

Development a Single Page Application(SPA) using the **MEAN stack (MongoDB, Express.js, Angular, Node.js)** to simulate the trading of second-hand mobile devices.

The system features a dual-interface architecture for seamless user transactions and robust administrative management, all built on **asynchronous MVC** design principles.

MVC Architecture

Model (Data)

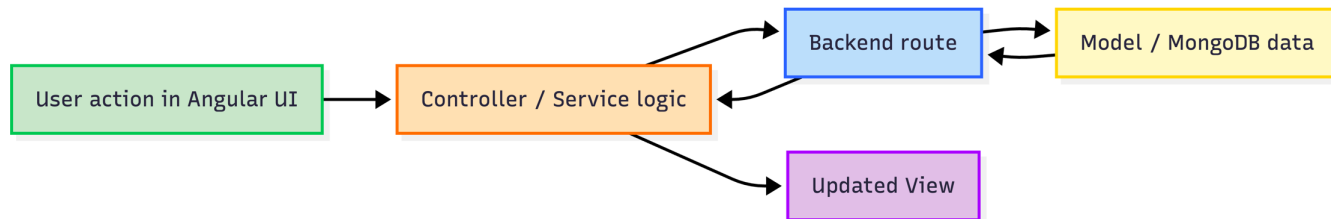
- Defines what data our system stores and the rules for that data.
- For example, [Phone Listing](#) stores title, brand, price, stock, reviews, and rating statistics, while User stores personal information, cart, and wishlist.

View (UI)

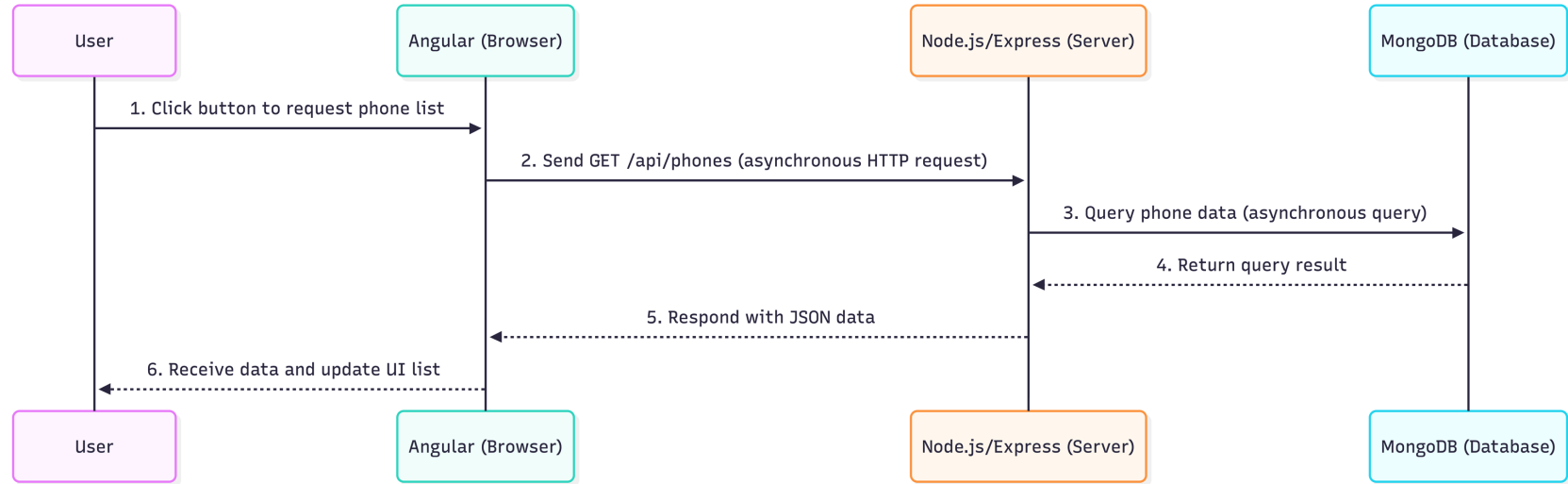
- The presentation layer. For example, Angular templates display product images, titles, prices, ratings, and stock information to the user.

Controller (Logic)

- Handles the application logic. For example, when a user requests phone listings, the route in [phoneRoutes.ts](#) receives the request, applies filtering or sorting, gets data from the model, and sends the result back to the frontend.



System Interaction Sequence Diagram



User Verification & Activation

Secure Password Storage

- **bcrypt Hashing:** Automatically generates a unique salt per user to compute a secure cryptographic hash, which is stored directly in MongoDB.

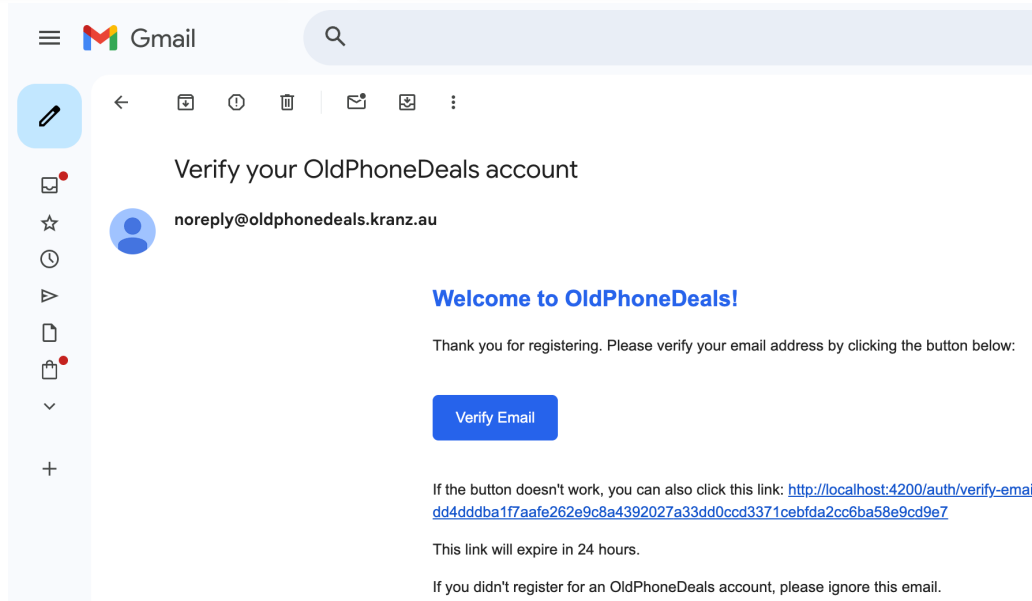
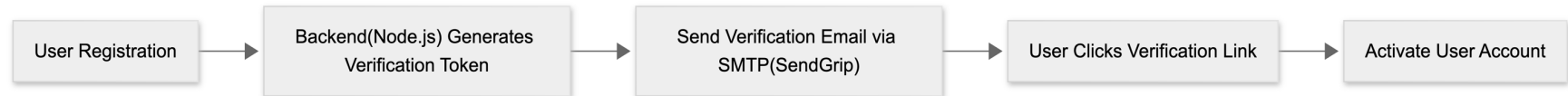
Time-Sensitive Token Generation

- **JWT:** Utilizes `jsonwebtoken` to issue verification tokens embedding the User ID, enforced with a 24-hour expiration window.

Token Validation & Route Protection

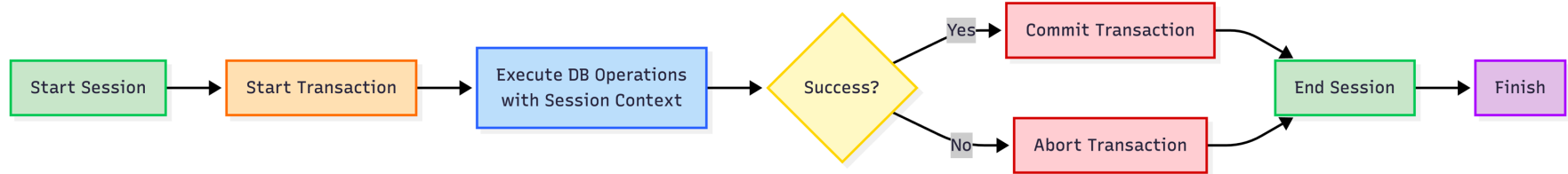
- **Signature Verification:** Backend executes `jwt.verify()` to validate the signature and expiration. Transitioning the user status from Pending to Active when success.
- **Auth Middleware:** Restricts sensitive routes by intercepting authentication attempts from pending accounts, blocking unauthorized access with a **403 Forbidden error**.

User Verification & Activation Implementation



Transaction Atomicity & ACID Compliance

Mongoose Transaction Lifecycle Management



Database operations during transaction

- **Order Creation:** Inserts a new transaction record into the orders collection.
- **Stock Decrement:** Deducts purchased quantities from `stock_quantity` field in listings collection.
- **System Notification:** Generates a real-time alert entry for administrative monitoring.
- **User Cart Clearance:** Clear cart array nested within the target user's document in `userscollection`.
- **Audit Trail Logging:** Commits a record to the sales and activity history logs.

Sold Out Soon & Best Sellers Implementation

OldPhoneDeals[Profile](#)[Wishlist](#)[Sign Out](#)

Sold out soon



4G Huawei Honor 6 Plus PE-UL00 F...
\$127.95
☆☆☆☆☆ (0.0)
0 reviews
Only 1 left!



Apple iPhone 3GS A1303 16GB GS...
\$39.00
☆☆☆☆☆ (2.5)
2 reviews
Only 1 left!



GreatCall Samsung Jitterbug Touch...
\$149.99
☆☆☆☆☆ (3.7)
9 reviews
Only 2 left!



Samsung A777 Unlocked Phone wit...
\$59.99
☆☆☆☆☆ (2.5)
6 reviews
Only 2 left!



Microsoft Lumia 650 (RM-1154) 16...
\$127.95
☆☆☆☆☆ (5.0)
1 review
Only 2 left!

Best sellers



Sam Galaxy J7(2016) Dual J7108 4...
\$231.00
☆☆☆☆☆ (5.0)
3 reviews



AT&T Motorola RAZR V3xx No Cont...
\$179.10
☆☆☆☆☆ (4.7)
3 reviews



'ATT Wireless LG CU720 Shine for i...
\$25.99
☆☆☆☆☆ (4.7)
3 reviews



"CLEAR CLEAN ESN" Sprint EPIC 4...
\$199.99
☆☆☆☆☆ (4.7)
3 reviews



Apple A1533 Unlocked iPhone 5S S...
\$115.00
☆☆☆☆☆ (4.4)
8 reviews

Homepage Design - Sold Out Soon Pipeline

"Sold Out Soon" Calculation

- Fetched products with stock greater than 0
- Sorts by stock in ascending order
- Returns the 5 products with the lowest remaining stock

```
fetchSoldOutSoon(): void {  
  this.isLoadingSoldOutSoon = true;  
  this.soldOutSoonError = null;  
  const filters: PhoneListingFilters = {  
    sortBy: 'stock',  
    order: 'asc',  
    limit: 5,  
    minStock: 1,  
    state: IsDisabledState.ENABLED,  
  };  
};
```

Homepage Design - Best Sellers Pipeline

"Best Sellers" Calculation

- Fetched products with available stock and at least 2 ratings
- Sorts by the stored `averageRating` in descending order
- Hidden reviews are still included when calculating `averageRating`

```
fetchBestSellers(): void {  
  this.isLoadingBestSellers = true;  
  this.bestSellerError = null;  
  const filters: PhoneListingFilters = {  
    sortBy: 'rating',  
    order: 'desc',  
    limit: 5,  
    minStock: 1,  
    minNumRatings: 2,  
    state: IsDisabledState.ENABLED,  
  };  
};
```

Performance Optimization - N+1 Queries

N+1 Query Prevention

- Uses Mongoose `populate()` to load sellers and review authors together
- Avoids querying related data separately for each product or review

```
const query = PhoneListingModel.find(filter)
  .populate('seller', 'firstname lastname')
  .populate('reviews.reviewer', 'firstname lastname')
  .sort(sortObject)
  .skip(skip)
  .limit(limitNumber);
```

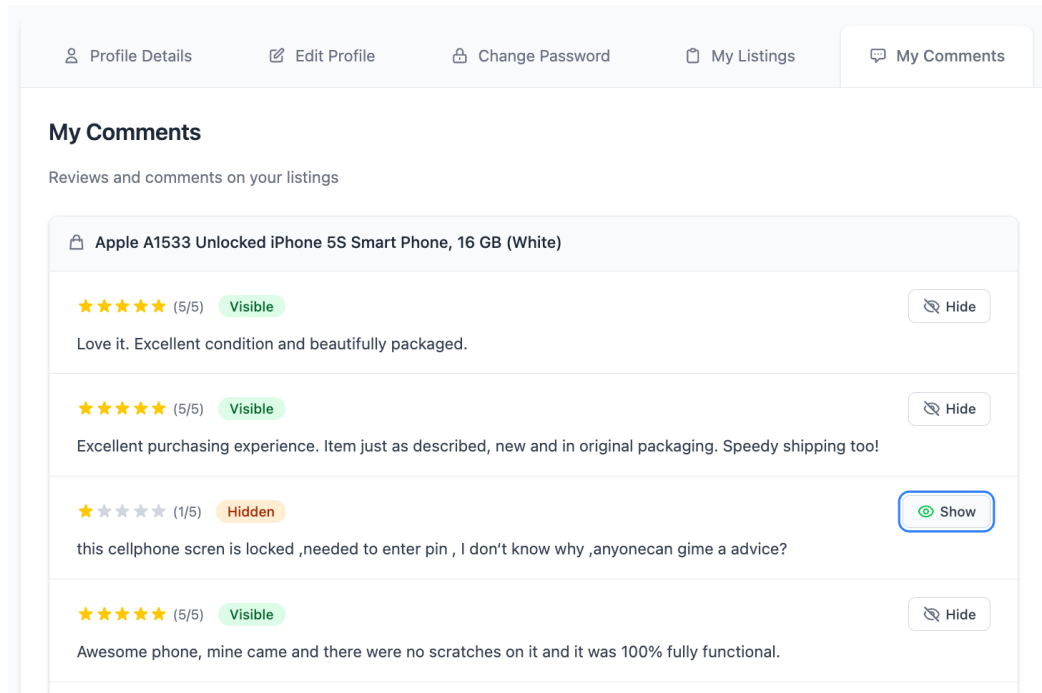
Role-Based Access Control for Comments

Layered Middleware Pipeline (Granular Access Control)

- 1. validateToken (Authentication Layer):** Verify signature of `JWT` from request header. If token is valid, it decodes the payload (incl. `userId` and `role`) and attaches it to the `req` object.
- 2. checkReviewPermissions (Authorization Layer):** Handle domain-specific business logic. For comments, it fetches the relevant review and product from the database. Then, it compares the current `user's ID` and role (`req.user`) against the data to determine their permissions.

Comments Implementation - Seller Interface

Listing owner can Show or Hide the comments below his/her phone listings



Comments Implementation - Role-Based Control

☐ Make this review private (only visible to you and the seller)

Submit Review

Peter Martinez
★★★★★
Love it. Excellent condition and beautifully packaged.

Marc Weekly
★★★★★
Excellent purchasing experience. Item just as described, new and in original packaging. Speedy shipping too!

Carla Manalili
★★★★★
this cellphone screen is locked ,needed to enter pin , I don't know why ,anyonecan gimme a advice?

Load More Reviews

Listing owner perspective

☐ Make this review private (only visible to you and the seller)

Submit Review

Peter Martinez
★★★★★
Love it. Excellent condition and beautifully packaged.

Marc Weekly
★★★★★
Excellent purchasing experience. Item just as described, new and in original packaging. Speedy shipping too!

Kenneth Jones
★★★★★
Awesome phone, mine came and there were no scratches on it and it was 100% fully functional.

Load More Reviews

Non listing owner (e.g. buyer) perspective

Session Expiration Warning Mechanism

Warning Mechanism

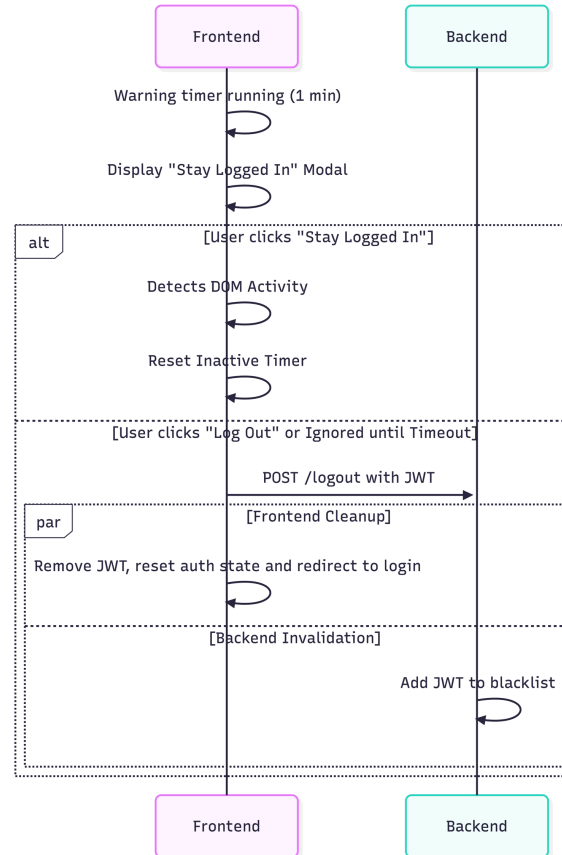
When admin login, Angular `AdminAuthService` starts two timers to monitor session validity.

1. **Main Inactivity Timer:** Set to the JWT's full lifetime (1 minute).
2. **Warning Timer:** Set to trigger 30 second before expiration.

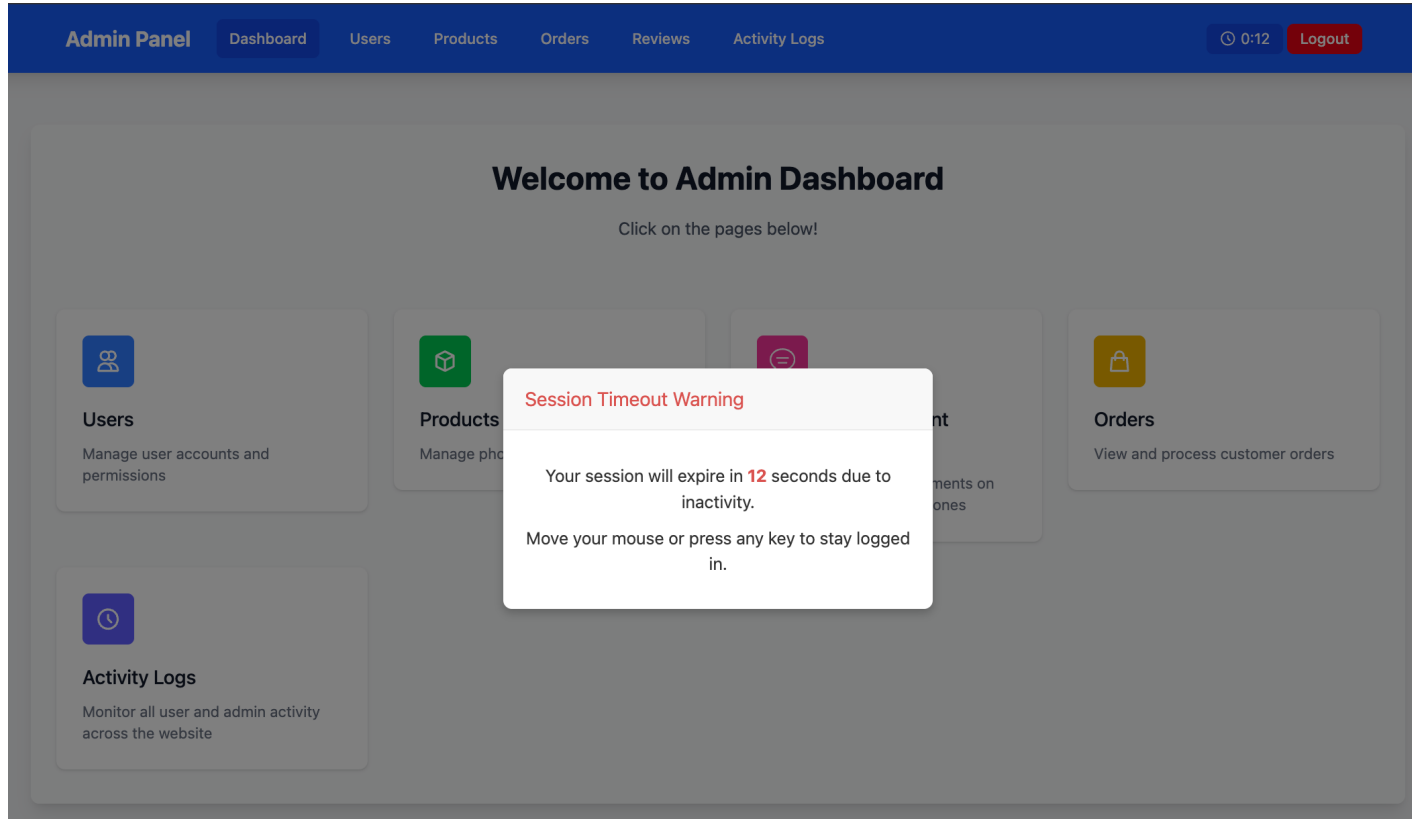
Timer Reset on User Activity

- Service listens for global `DOM events` like mousemove, keypress, and scrolling.
- Any detected user activity calls a `resetInactiveTimer()`, which clears existing timers and restarts.

Session Expiration Lifecycle Sequence



Session Expiration Warning Implementation



Technical Highlights - Architecture & Auth



SPA Principle

- Home / Search / Item views load within the same page **without reload**
- All client-server communications are **asynchronous**

MVC Architecture

- Clear separation of **Model (data)**, **View (UI)**, and **Controller (logic)** for maintainability and scalability

User Information Security

- **bcrypt hashing** and **salted**
- email must be **verified before login**
- password can **reset via email link**

Technical Highlights - Data Integrity & Security



Database Architecture & Transaction Integrity

- **MongoDB** dynamic schema (**NoSQL**)
- **ACID-Compliant Transactions**

Performance & Security

- **Proper indexing** and Mitigated **N+1 Query Bottlenecks**
- Granular **Role-Based Access Control (RBAC)**
- Synchronized **Session Lifecycle & Revocation**
- Prevented **XSS** attacks
 - Angular template interpolation automatically escapes user-generated content, e.g. `{{ userReview.comment }}`

Thank you!



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